

Determining the Best Agile SDLC for Bangladesh's Software Industry

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Abstract—Agile methodologies start possessing popularity in the last decade. In Bangladesh this was the time when software development starts to emerge as a promising industry. But most of the Software development organizations, including small outsourcing oriented firms in Bangladesh follow Waterfall method as their development process model. Recently, some of the firms have started following agile methodologies for testing basis but they are not willing to transform their whole development process from Waterfall to agile. This study focuses on the scopes and the benefits of following agile methodologies instead of Waterfall model. This study also highlights the best fit methodology to follow for the emerging software development industry in context of Bangladesh.

Index Terms— Agile methodologies, Scrum, SDLC, Waterfall Model, XP.

I. INTRODUCTION

THE idea of software development life cycle starts when Dr. Winston Royce described a model to manage large software development projects in 1970. In his model he suggested to do the design and related research before starting programming. After some days his model started to be called Waterfall model and becomes the most followed SDLC. But to cope with the real constantly changing business world, some new methodologies has been introduced. These methodologies having some differences have one thing in common, all of these suggest to break the requirement into small stories so that a change cannot affect the whole system. And this was the starting of module based software development. Agile methodologies let the followers to break down a big project to parts and make it more manageable. These methodologies ensure that every member of the team is doing his best and keep everyone of the team informed about the status of others. In Bangladesh Waterfall method is followed by most of the software development organization, regardless of being big or small. But these practice causes a lack of flexibility in requirement changing. Clients always feel free to change requirement whenever he wants to and the developer team are not willing to entertain client's changes most of the times. Following Agile method can increase flexibility in changing requirement, can decrease development time and complexity. But the most important and fabulous thing about Agile methodologies is it increases the spirit and mutual understanding among team members. This study

comes out with the result of following Agile methodologies instead of following Waterfall method and describes the benefits of it for Bangladeshi software firms.

II. BACKGROUND STUDY

A. Waterfall Method

Chronological paradigm of software designing progression is known as waterfall model perceiving the development method as water falling gradually downhill. This method implies the significance of carrying tasks sequentially in a precise progressive order. This is the primogenital method followed profoundly throughout the software industry. The five phases of Waterfall model are shown in Fig. 1. The team must start with phase one which is Communication and got to end up at phase development. Through the entire expedition of development life cycle, one cannot jump to any other phases prior or after the current phase as desired. Following the sequence in order which minimizes the risk factors to a great extent.

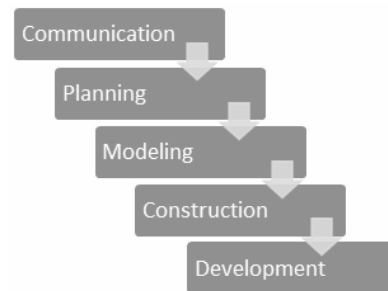


Fig. 1. Waterfall Model

B. Agile Methods

Agile methods' main idea is to break projects into small requirements and completing those requirements one after another. Small requirements are divided into short stories and completion of all the stories completes a requirement. Scrum and Extreme Programming (also known as XP) are two of the most practiced agile methodologies. This section will highlight Scrum and XP methods briefly.

1) SCRUM Methodology

When following the Scrum, a project requirement is divided into small stories and with some stories a sprint is organized. A sprint is a short duration of time to complete a specific task. If any change in requirement occurs then it possibly affects one or two stories, not all of them. So, the development team

faces less complexity. Every story is designed in such a way that when implementing a story, a developer does not need to think about other stories. The process of Sprint management is shown in Fig. 2. Scrum is thought to be a framework to manage projects but later it has been adopted as a software development process. The difference between Scrum and other methods is that when following Scrum it is assumed that analysis, design, and development processes in a Sprint are not stable. A process is used to manage the risk and unpredictability and ensuring flexibility, responsiveness, and reliability. Scrum is fully responsive to changes and adjusts the project in every sprint. Scrum methodology needs the teams working on a project to sit together every day which is known as the 'Daily Meeting'. A 'Sprint Planning Meeting' and 'Sprint Review Meeting' are also arranged at the starting of the sprint and at the end of a sprint respectively to keep every member of the project teams and the client updated about the project status. The Scrum practices [1] are listed below.

- Product Backlog
- Effort Estimation
- Sprint
- Daily Meeting
- Sprint Planning Meeting
- Spring Backlog
- Sprint Review Meeting
- Sprint Burn Down Chart

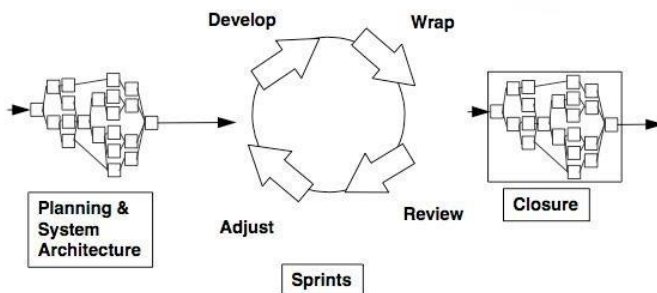


Fig. 2. SCRUM Development Process

2) Extreme Programming (XP) Methodology

XP is one of the widely used method where initial requirements are likely to change in a large scale [7]. XP depends on five values to complete a project; communication, simplicity, feedback, courage and respect. It has some unique practices to ensure quality software development such as test-first programming, pair programming, incremental design, continuous integration etc. This practices and some principles such as humanity, quality, mutual benefit, improvement, diversity, failure, flow, opportunity, redundancy, etc. are used to bridge the gap between values. Like Scrum, XP also divides requirement into stories but instead completing a spring within a week and arranging multiple meetings, XP summarizes the whole project at first weekday. Application definition and requirement are redefined continuously every week. As a

result XP deliver rapid system solution with low rate of bugs [5]. The XP practices are listed in the table below.

TABLE I
EXTREME PROGRAMMING PRACTICES [8]

Sit Together	Incremental Design
Whole Team	Real Customer Involvement
Informative Workspace	Incremental Deployment
Energized Work	Team Continuity
Pair Programming	Shrinking Teams
Stories (User Stories)	Root Cause Analysis
Weekly Cycle	Shared Code
Quarterly Cycle	Code And Tests
Slack	Single Code Base
Ten Minute Build	Daily Deployment
Continuous Integration	Negotiated Scope Contract
Test First Programming	Pay Per Use

III. COMPARISON BETWEEN SCRUM AND XP

Scrum and Extreme Programming (XP) are very much alike in their working procedure. In fact, if someone just switched his/her team doing one of these processes, it will be quite tough for him/her to find out which team he/she is into; a Scrum team or an XP team. The border line between the two processes are very thin, but the differences are important. There are four main differences between Scrum and XP [6].

In Scrum, the teams work in cycles (the sprints) that are usually from two weeks to one month long. Sometimes, a sprint is divided into shorter sprints that can be completed within a day or two [1] [2]. Whereas, in XP, teams usually work in iterations much smaller in duration comparing to a regular Scrum cycle, usually one or two weeks.

Scrum do not allow any change in their sprints after the planning meeting is done and a commitment is made to deliver one or more product backlog items [2]. If there is any change that must be done, it is done at the end of the sprint. XP teams are much more flexible in context of changing within their iteration. If a particular feature is not started yet, it can be changed. Usually a new feature of equivalent size is swapped in exchange for the un-started feature.

XP teams work in a priority order that is quite strict. Customer defines the priority of the items and the team follows that order. On the other hand, the Scrum product owner (similar to XP customer) prioritizes the items but the team can choose the sequence to work on. It is not mandatory for a Scrum team to start working from the highest-priority item [1] [4]. It is quite obvious for the team to start from one of the lower prioritized items. Sometimes one of the high priority items may not be a good fit for the sprint being planned—maybe a team member or a key requirement is unavailable right that moment. Or may be a lower priority item (let's assume #5) which is not dependent on a higher priority item (let's assume #2) can be developed earlier whereas #2 needs some more time to work on it.

There is no definite engineering practice defined for Scrum but it is there for XP like test-driven development, pair programming, simple design, automated testing, refactoring etc.

It would be a tough choice to switch to an agile method for

development because it is difficult to adopt a new process [4]. Both Scrum and XP offers quite similar advantages. A better performance may be achieved by not to stick around a single practice but combining both the practices. It is observed that starting with the Scrum and then switching to any of the XP practices can obtain a better result [6].

IV. DETERMINING THE BEST METHODOLOGY

The main goal of this study is to determine the best agile methodology for software industries in Bangladesh. Most of the software firms of Bangladesh works for foreign clients. Sometimes the clients doesn't provide the whole requirement at the beginning and the team are also not clear about the final product because of their differences and lacking in foreign language. This situation requires a process that can develop a system properly based on provided information while keeping scope for continuous change in the requirement. Moreover the new and small scale firms work on small and low budget projects. At most of the cases the estimated time of a project is less than 3 months and resources are very limited. They need a proficient and well managed methodology that can help the team to develop the system within limited time frame and resource and also deliver an efficient and satisfactory product to please the client to ensure winning future projects from foreign clients.

In order to compare two methodologies, a set of attributes should be defined first that will serve as benchmark. Fernandes and Almida [3] performed a comparison between agile methods using a set of four attribute extracted from the IEEE's Software Engineering Body of Knowledge (SWEBOK) Knowledge Areas (KA) and all the other principles defined in the Agile Manifesto [5] were encapsulated as the 5th attribute. Their objective was analyzing the practices proposed by each method to evaluate the coverage degree for the KAs and the agility of the methods. The KAs selected were as follows:

- i. Software Requirements
- ii. Software Construction
- iii. Software Testing
- iv. Software Engineering Management

For each attribute, the authors utilized a qualitative classification system, the criteria labels are defined in the table below.

TABLE II
FERNANDES AND ALMEIDA'S COMPARISON CRITERIA [3]

Label	Meaning	Details
NS	Not Satisfied	None of the proposed practices or concepts of the method support the attribute or principle.
PS	Partially Satisfied	The proposed practices or concepts of the method support the sub-attribute or principle, although some of its aspects are not considered.
AS	Adequately Satisfied	The proposed practices or concepts of the method entirely support the sub-attribute or principle.

Based on Fernandes and Almeida's criterions we have studied the Bangladeshi Software industry to determine the best methodology for this industry's most of the firms.

TABLE III
SATISFACTORY PERCENTAGE FOR DIFFERENT ATTRIBUTE

Attributes	XP			Scrum		
	AS	PS	NS	AS	PS	NS
Software Requirements	60%	40%	0%	60%	20%	20%
Construction of Software	75%	25%	0%	0%	0%	100%
Software Testing	100%	0%	0%	0%	0%	100%
Soft Engineering Management	80%	0%	20%	80%	20%	0%
Agile Principles – Proposed Practices Relation	75%	8%	17%	50%	17%	33%

Our results show that XP performed better on the attributes Software Requirement, Construction of Software and Software Testing while Scrum improved on Software Engineering Management. The practice of XP that suggests software teams should have a close relationship with the customer, had a positive impact in the Software Requirements attribute; other practices such as Test First and Pair programming reflected good scores as well. Regarding the "agility" of the methodologies, it is not surprising that XP had a better score than Scrum because XP has influenced the principles that form the Agile Manifesto [3].

Finally we conclude different projects demand different requirements and the selection of a methodology depends on the context of development while keeping the environment and resources in mind.

V. CONCLUSION

The word "Agile" has become a synonym for "modern software practice that work" [6]. It is proven in research as well as in the industry that agile methodologies significantly increase quality of software. Also agile processes produce a software in an incremental approach which can reduce the risk of failure and ensure customer satisfaction [4]. However all software projects are different and a careful step should be taken in order to determine the best fit methodology.

For the software firms of Bangladesh XP could be the best agile methodology to be used at most of the projects. Almost all of the firms work on outsourced projects and they need to keep their clients informed about everything related to that project. The firms could not afford enough employee for a project because of low budget and also they need to utilize their resources like time and man power at its best. And last but not the least, they need to ensure the quality of the end product. So XP is the best fit methodology to follow by ensuring changes based on project needs and resource limitation [8].

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